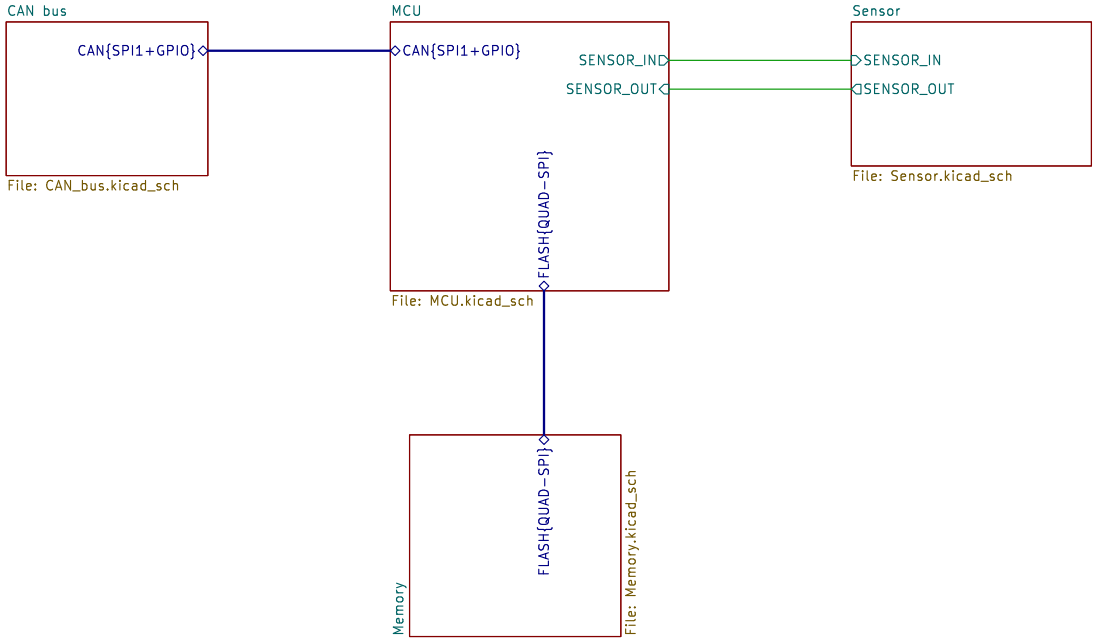


H101
MountingHole

H102
MountingHole

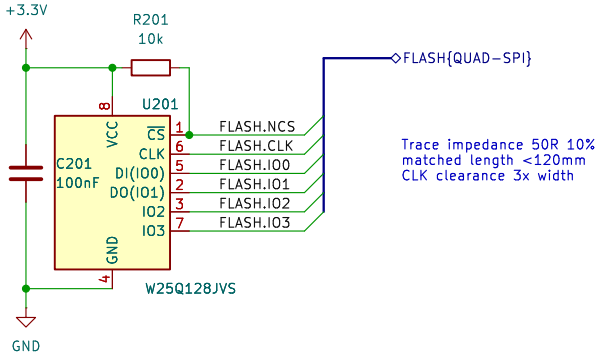
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

Sheet: /
File: BB_nosepoke_V1.kicad_sch

Title: BB_nosepoke

Size: A4	Date: 2024-03-21	**Rev: v1**
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Quad-SPI FLASH NOR Memory

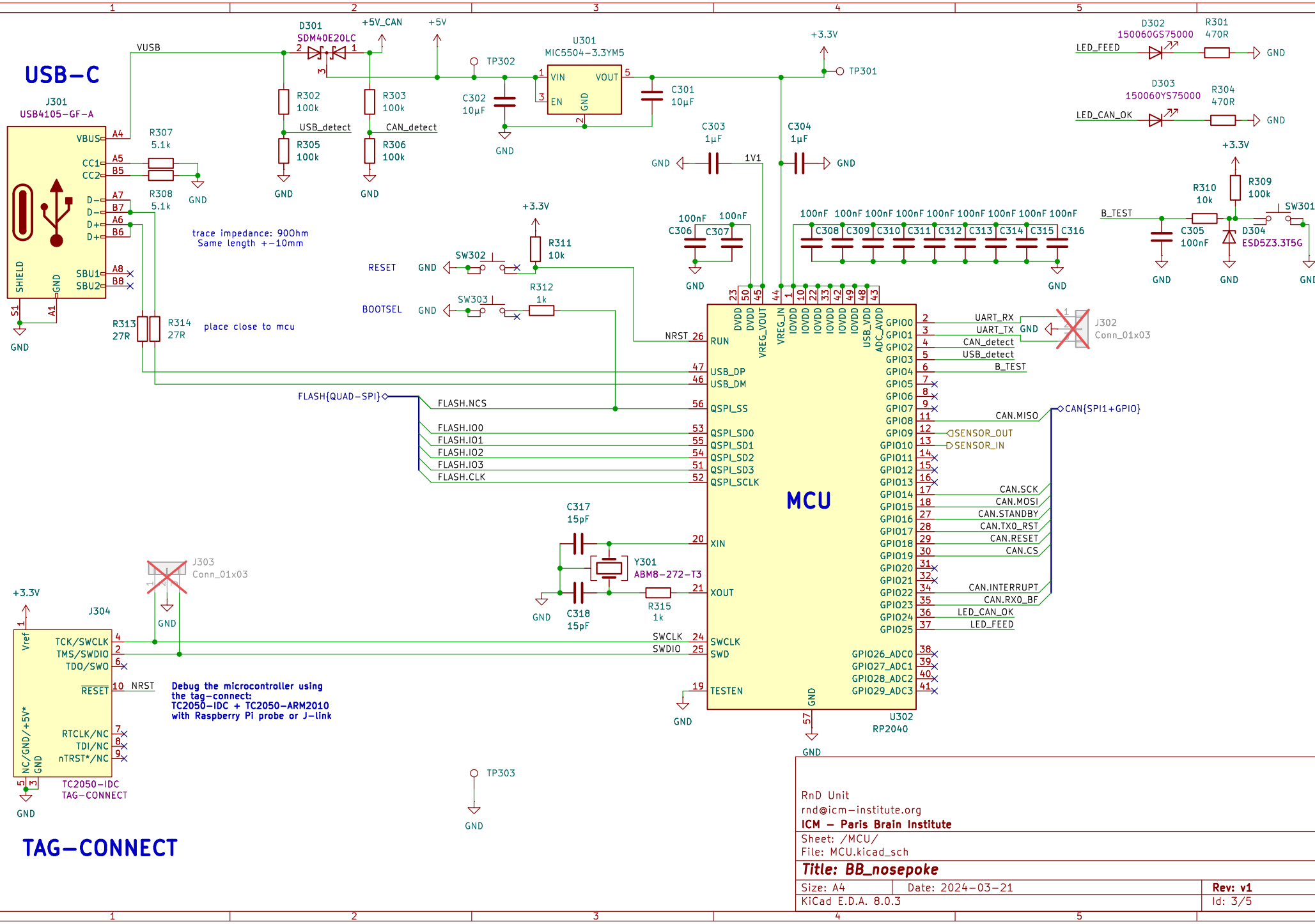


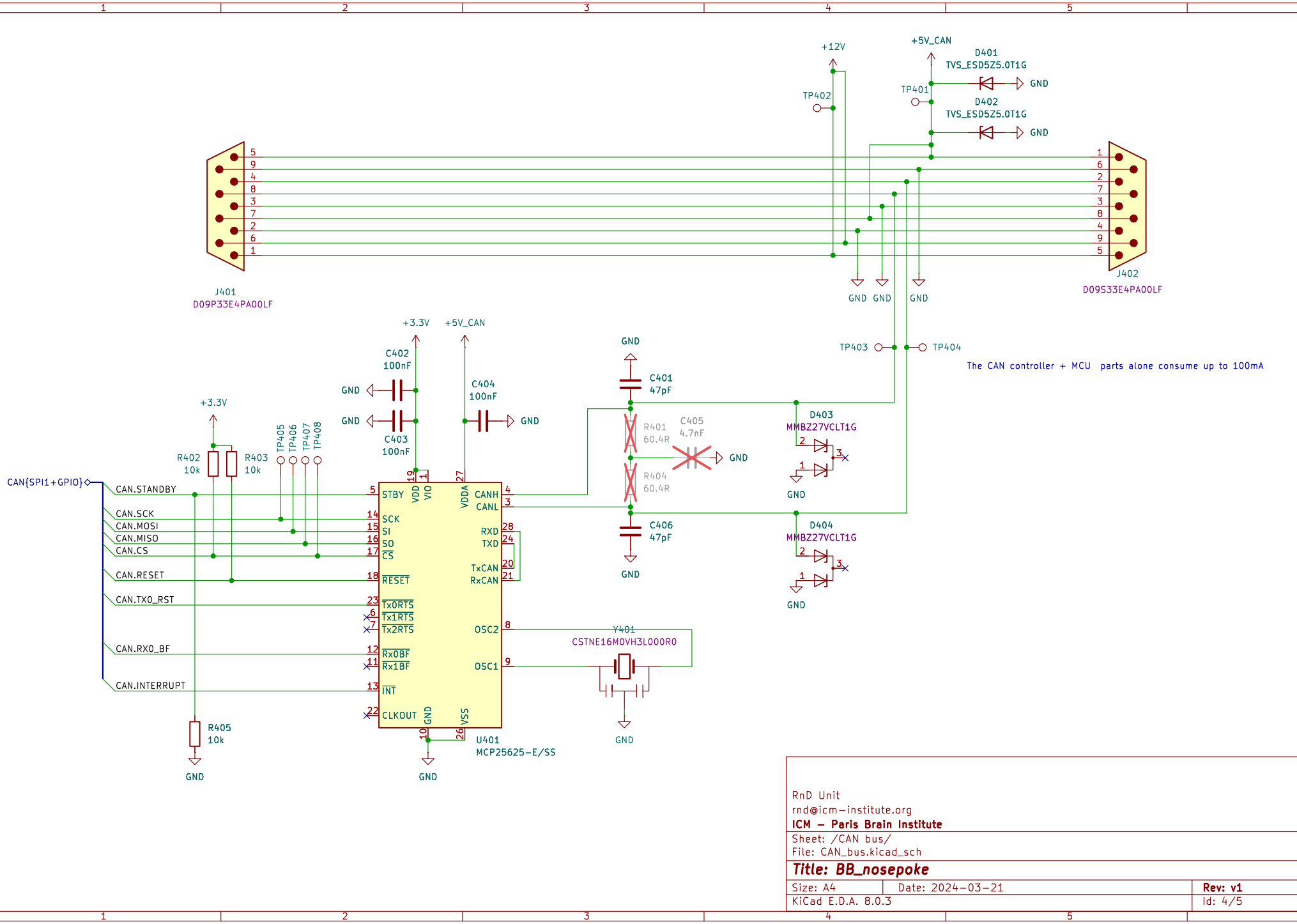
RnD Unit
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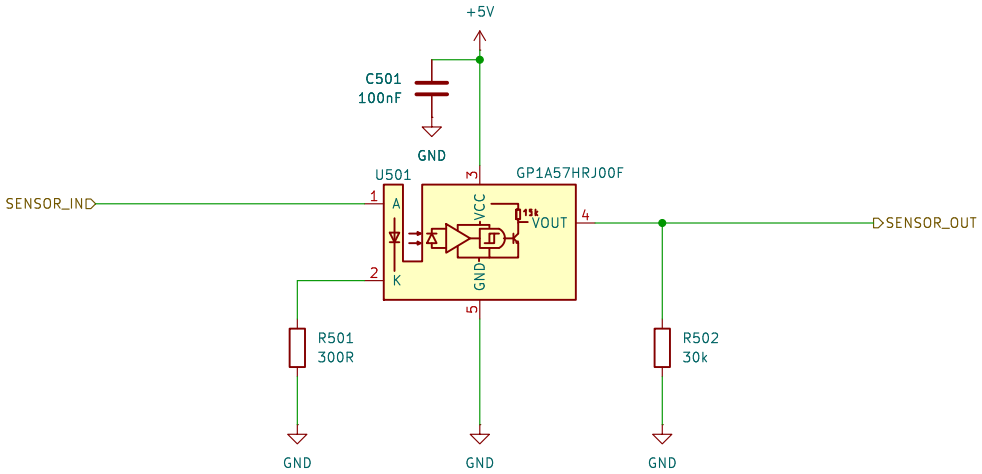
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File: Memory.kicad_sch

Title: BB_nosepoke

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 8.0.3		Id: 2/5


USB-C
J301
USB4105-GF-A
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
VUSB
R307 5.1k
R308 5.1k
R313 27R
R314 27R
place close to mcu
trace impedance: 90Ohm
Same length +-10mm
D301
SDM40E20LC
+5V_CAN
+5V
R302 100k
R303 100k
R305 100k
R306 100k
USB_detect
CAN_detect
U301
MIC5504-3.3YM5
VIN
VOUT
EN
GND
TP302
C302 10µF
C301 10µF
+3.3V
TP301
C303 1µF
C304 1µF
1V1
D302
150060GS75000
R301 470R
LED_FEED
D303
150060YS75000
R304 470R
LED_CAN_OK
R309 100k
R310 10k
B_TEST
SW301
C305 100nF
D304
ESD5Z3.3T5G
C306 C307 100nF
C308 C309 C310 C311 C312 C313 C314 C315 C316 100nF
SW302
SW303
RESET
BOOTSEL
R311 10k
R312 1k
MCU
RP2040
U302
DVDD
VREG_VOUT
VREG_IN
IOVDD
USB_VDD
ADC_AVDD
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
FLASH{QUAD-SPI}
FLASH.NCS
FLASH.IO0
FLASH.IO1
FLASH.IO2
FLASH.IO3
FLASH.CLK
C317 15pF
C318 15pF
Y301
ABM8-272-T3
R315 1k
UART_RX
UART_TX
CAN_detect
USB_detect
B_TEST
J302
Conn_01x03
CAN.MISO
CAN.SCK
CAN.MOSI
CAN.STANDBY
CAN.TXO_RST
CAN.RESET
CAN.CS
CAN.INTERRUPT
CAN.RXO_BF
LED_CAN_OK
LED_FEED
SENSOR_OUT
SENSOR_IN
CAN{SPI1+GPIO}
J303
Conn_01x03
J304
Vref
TCK/SWCLK
TMS/SWDIO
TDO/SWO
RESET
NRST
RTCLK/NC
TDI/NC
nTRST*/NC
NC/GND/+5V*
TC2050-IDC
TAG-CONNECT
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
TP303
TAG-CONNECT
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_nosepoke
Size: A4
Date: 2024-03-21
Rev: v1
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+12V
+5V_CAN
D401
TVS_ESD5Z5.0T1G
GND
TP401
D402
TVS_ESD5Z5.0T1G
GND
TP402
J401
D09P33E4PA00LF
J402
D09S33E4PA00LF
GND GND GND
TP403
TP404
The CAN controller + MCU parts alone consume up to 100mA
+3.3V
+5V_CAN
C402
100nF
GND
C403
100nF
GND
C404
100nF
GND
GND
C401
47pF
R401
60.4R
C405
4.7nF
GND
R404
60.4R
C406
47pF
GND
D403
MMBZ27VCLT1G
GND
D404
MMBZ27VCLT1G
GND
+3.3V
R402
10k
R403
10k
TP405
TP406
TP407
TP408
CAN{SPI1+GPIO}
CAN.STANDBY
CAN.SCK
CAN.MOSI
CAN.MISO
CAN.CS
CAN.RESET
CAN.TX0_RST
CAN.RX0_BF
CAN.INTERRUPT
STBY
SCK
SI
SO
CS
RESET
TX0RTS
TX1RTS
TX2RTS
RX0BF
RX1BF
INT
CLKOUT
VDD
VIO
GND
VSS
VDDA
CANH
CANL
RXD
TXD
TxCAN
RxCAN
OSC2
OSC1
Y401
CSTNE16M0VH3L000R0
GND
R405
10k
GND
GND
U401
MCP25625-E/SS
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /CAN bus/
File: CAN_bus.kicad_sch
Title: BB_nosepoke
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 8.0.3
Id: 4/5



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Sheet: /Sensor/
File: Sensor.kicad_sch

Title: BB_nosepoke

Size: A4	Date: 2024-03-21	**Rev: v1**
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